(3-d) (16) (17)

Q1) If the transit time for the signal to reach the center of the house from the CCTV is sufficiently small compared to the time period of the signal itself, we can use the lumped model approximation.

$\lambda = 0.3m \quad 3\times10^8 = f \times 0.3 \Rightarrow f = 10^9 \Rightarrow T = 1ns$

$t = \ell/v = \ell/2\times10^8 \qquad \ell/2\times10^8 << 1ns \Rightarrow \ell/2\times10^8 \times 10 < 1\times10^{-9} \Rightarrow \ell \leq 2cm$

4 cm diameter house

Q2) V → voltage across side branch ; ΔV – voltage across main branch

$I+\Delta I$ → current flowing out of main

I → current flowing from side

model (B) : $\Delta I = (G\Delta z + j\omega C\Delta z)V$ (ohm's law in side)

$\Delta V = (R\Delta z + j\omega L\Delta z)(I+\Delta I)$ (ohm's law in main)

$$\Rightarrow \frac{dI}{dz} = (G+j\omega C)V \qquad \frac{dV}{dz} = (R+j\omega L)I$$

$$\Rightarrow \frac{d^2V}{dz^2} = \gamma_b^2 V \qquad \gamma_b = \sqrt{(R+j\omega L)(G+j\omega C)}$$

model (A): Main I_m ⇒ capacitor current = $I_m - I$,

conductance current = $I + \Delta I - I_m$

$$V = \frac{2}{j\omega C\Delta z}(I_m - I) \qquad I + \Delta I - I_m = (V+\Delta V)\frac{G\Delta z}{2} \qquad \Delta V = (R\Delta z + j\omega L\Delta z)I_m \approx (R+j\omega L)\Delta z\, I$$

$$\Rightarrow I_m - I = \Delta I - \frac{VG\Delta z}{2}$$

$$\frac{dI}{dz} = \frac{G+j\omega C}{2}V \qquad \Rightarrow V = \frac{2}{j\omega C}\left(\frac{dI}{dz} - \frac{VG}{2}\right) \qquad \Rightarrow \frac{dV}{dz} = (R+j\omega L)I$$

$$\Rightarrow \frac{d^2V}{dz^2} = \gamma_a^2 I \Rightarrow \gamma_a = \sqrt{(R+j\omega L)(G+j\omega C)}/\sqrt{2} \rightarrow \gamma_a/\gamma_b = 1/\sqrt{2}$$

Q3) $R = 0 \; ; \; G = 0 \; ; \; L = 245\times10^{-9} \; ; \; C = 200\times10^{-12}$

(a) z_0 → purely resistive ; $\sqrt{L/C} = \sqrt{\frac{245000}{200}} = 35\Omega$

(b) $z_{in} = 35 \left\{ \frac{100 + 35 \tanh(\gamma L)}{35 + 100 \tanh(\gamma L} \right\}$ (always complex)

(c) $\frac{100 - 35}{100 + 35} = 0.481$ $\quad S = \frac{1 + \Gamma_L}{1 - \Gamma_L}$ $\quad \frac{1 + 65/135}{1 - 65/135} = 2.84$

(d) Better TL $\Rightarrow$ Lower S

(Q4) $\omega = \frac{5 \times 10^8 \text{ Hz}}{2\pi}$ $\quad z_0 = 80\Omega$ $\quad \alpha = 0.04$ NP/m $\quad \beta = 1.5$ rad/m

$\alpha + j\beta = \sqrt{(R + j\omega L)(G + j\omega C)}$

$z_0 = \sqrt{\frac{R_0 + j\omega L}{G + j\omega C}} \Rightarrow (\alpha + j\beta) z_0 = R + j\omega L$

$\Rightarrow \alpha = R/z_0$ i $\beta = \omega L / z_0$

$R = 3.2\Omega/m$

$L = 3.8 \times 10^{-9}$ H/m

$G = 0.3$ mU/m

$C = 5.96 \times 10^{-12}$ F/m

(Q3) $z_{in} = 300\Omega$ $\quad z_L = 300\Omega$

$z_0 = 300\Omega$ $\quad v = 2.5 \times 10^8$ m/s

$V(z) = V_0^+ e^{-\gamma z} + V_0^- e^{\gamma z}$

$V(x=0) = V_0^+ + V_0^- = V_0^+$

$V(x=l) = V_0^+ e^{-\gamma l} + V_0^- e^{\gamma l} = V_0^+ e^{-\gamma l}$

$\Gamma = 0 \Rightarrow V_0^- = 0$ $\quad$ phase difference $= \gamma l = \beta l$

$\phi = 8\pi/5$

$\lambda = \frac{2.5 \times 10^8}{100 \times 10^6} = 2.5$ m $\quad$ i $\quad \beta = 2\pi/2.5 = 4\pi/5$

(b) $\Gamma = 0$, $VSWR = 1$

(c) $Z_L = 150\Omega$

$\Rightarrow \Gamma = \frac{-150}{450} = -1/3$; $VSWR = 2$

(d) before: $Z_{in} = Z_0 = 300\Omega \Rightarrow Z_{look\text{-}in} = 600\Omega$

after: $Z_0 = 300\Omega$ $Z_L = 150\Omega$ $\beta l = 8\pi/5$

$$Z_{in} = 300\left[\frac{1 + 2\tan(8\pi/5)\,j}{2 + \tan(8\pi/5)\,j}\right] = 466 - 205j$$

$$\Rightarrow Z_{look\text{-}in} = 766 - 205j$$

(Q6) $Z_0 = 600\Omega$ $Z_L = 150j\ \Omega$ $\beta = \omega/v_p = 2\pi$ $\boxed{\beta = \frac{2\pi}{\lambda}}$

$$Z_{in} = 600 \cdot \frac{600}{150j} = -2400j$$

(Q7) short circuited $\Rightarrow Z_L = 0 \Rightarrow \Gamma = -1 = \frac{V_0^-}{V_0^+}$ lossless $\alpha = 0$

$$V(l) = V_0^+ e^{j\beta l} + V_0^- e^{-j\beta l} = 2V_0^+ \sin(\beta l)$$

$\beta = \frac{20\pi}{3}$

$$2V_0^+ \sin\left(\frac{20\pi}{3} \times \frac{8}{10}\right) = 10 \Rightarrow 2V_0^+ \left(\frac{\sqrt{3}}{2}\right) = 10$$

$$\Rightarrow 2V_0^+ = 20/\sqrt{3}$$

$$I(0) = \frac{V_0^+ - V_0^-}{Z_0} = \frac{2V_0^+}{Z_0} = \frac{20}{\sqrt{3} \cdot 300} = \frac{1}{15\sqrt{3}}\ A$$

(Q8) $Z_0 = 50\Omega$ $Z_L = ?$ $S = 2.4 = \frac{1 - |\Gamma_L|}{1 + |\Gamma_L|} \Rightarrow |\Gamma_L| = 7/17$

$$V(l) = V_0^+ e^{j\beta l} + V_0^- e^{-j\beta l} = V_0^+ e^{j\beta l}\left[1 + |\Gamma_L| e^{j(\phi_L - 2\beta l)}\right]$$

ϕ_L → phase shift b/w incident voltage wave and reflected voltage wave on load.

$\phi_L - 2\beta l = 2K\pi$ $K = 0$, $\beta = \frac{2\pi}{\lambda}$, $l = \frac{\lambda}{8} \Rightarrow \phi_L = \pi/2$

$$\Rightarrow \Gamma_L = \frac{7}{17} e^{j\phi_L} = \frac{7j}{17}$$

$$z_L = 50\left[\frac{17+7j}{17-7j}\right]$$

$$\phi_L - 2\beta l = -\pi \Rightarrow 2\beta l = \frac{3\pi}{2} \Rightarrow l = \frac{3\lambda}{8}$$

(Q9) $Z_0 = 50\Omega \quad S = 5 \Rightarrow \frac{1+|\Gamma_L|}{1-|\Gamma_L|} = 5 \Rightarrow |\Gamma_L| = 2/3$

$\lambda/2 = 0.5 \Rightarrow \lambda = 1m \rightarrow \beta = 2\pi$ first maxima @ $\lambda/4$

$$V(l) = V_0^+ e^{j\beta l}\left[1 + |\Gamma_L| e^{j(\phi_L - 2\beta l)}\right]$$

$$\phi_L = 2\beta l = 2 \times 2\pi \times 1/4 = \pi \Rightarrow \Gamma_L = 2/3 e^{j\pi} = -2/3$$

$$-2/3 = \frac{z_L - 50}{z_L + 50} \quad -2z_L - 100 = 3z_L - 150 \Rightarrow z_L = 10$$

$$V_l = V_0^+\left[1 - 2/3\right] \Rightarrow V_0^+ = 300V \quad V_0^- = -200V$$

$$I_0^+ = 6A \quad I_0^- = -4A$$

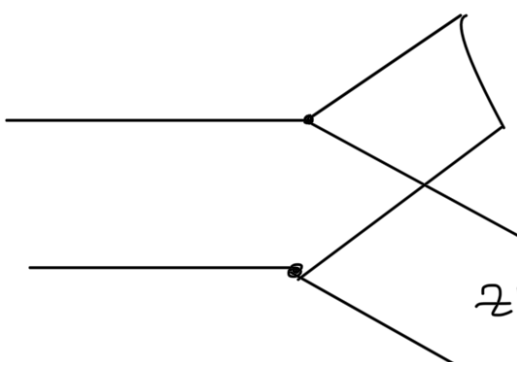
(Q10) $l = 500m \quad \phi_L = \frac{2\pi}{6} \quad \lambda = 150m$

$$2\pi/6 - 2\cdot 2\pi/150 \cdot l = 0 \Rightarrow l = 175/4\ m$$

$$\frac{175}{4} + (p-1)\frac{150}{2} \le 500$$

$$\Rightarrow 175 + (p-1)\cdot 300 \le 2000 \rightarrow (p-1) \le \frac{1825}{300} \Rightarrow 7 \text{ maximas}$$

(Q11)



$$z_{in(B)} = z_0 j \cdot \tan(\beta l) = jz_0$$

$$l = \lambda/8$$

$$z_{in(A)} = -j z_0 \tan(\beta l) = -jz_0$$

$$\Rightarrow (Z_L)_C = \infty \,,\ \Gamma_L = 1 \Rightarrow S = \infty$$

(Q12) $v = \omega/\beta \Rightarrow \beta = \frac{2\times10^9 \cdot 2\pi}{2\times10^8} = 20\pi = 2\pi/\lambda \Rightarrow \lambda = 1/10\ m$

$\alpha = 1.5/8.68$ NP/m

$\beta = 20\pi$ rad/m

for low loss lines, an open circuited quarter wave line transforms the open circuit into a short circuit

$$l_{sc} = \lambda/4, 3\lambda/4, \ldots = 1/40, 3/40 \ldots$$

$$l_{oc} = \lambda/2, \lambda, 3\lambda/2 \ldots = 1/20, 2/20, \ldots$$

for S.C $\quad z_{in} = \frac{z_0}{\alpha l_{sc}} = 11.57\,\Omega\times10^3$

for O.C $\quad z_{in} = \frac{z_0}{\alpha l_{oc}} = 5.79\,\Omega\times10^3$

Quality factor $= \frac{\beta}{2\alpha} = 181.8$

3-dB Bandwidth $= \frac{f_0}{Q} = \frac{2\times10^9}{181.8} = 11$ MHz

(Q13) $10\times\frac{50}{100} = 5 = -20\log\left(\frac{V_{load}}{V_{source}}\right) \Rightarrow V_l = 10^{3/4}$ V

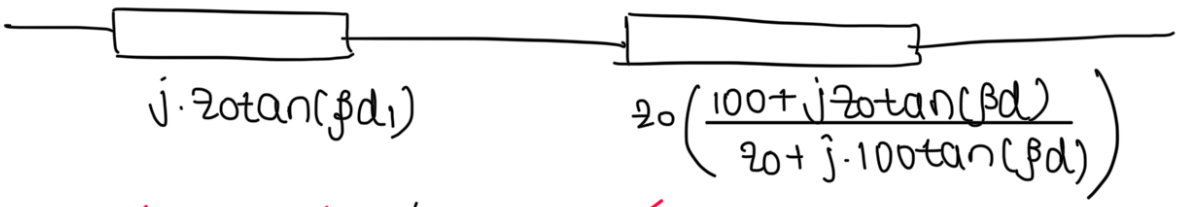
$\downarrow$ 10

$= -20\log\left(\frac{I_{load}}{I_{source}}\right) \Rightarrow I_l = 3\times10^{-1/4}$ A

$P = \frac{1}{2}\times 10^{3/4}\times 3\times10^{-1/4} = 1.5\times\sqrt{10} = 4.75$ W

(we can't if impedance isn't matching as there'll be reflections)

(Q14)



$$\Rightarrow z_0 = j \cdot z_0 \tan(\beta d_1) + z_0\left(\frac{100 + j z_0 \tan(\beta d)}{z_0 + j \cdot 100 \cdot \tan(\beta d)}\right)$$

$$1 = j \tan(\beta d_1) + \frac{100 z_0 (\sec^2(\beta d)) + j \tan(\beta d)\,[z_0^2 - 100^2]}{z_0^2 + 100^2 \tan^2 \beta d}$$

$$\frac{100 z_0 \sec^2(\beta d)}{z_0^2 + 10^4 \tan^2(\beta d)} = 1 \qquad \frac{(10^4 - z_0^2)\tan(\beta d)}{z_0^2 + 10^4 \tan^2 \beta d} = -\tan(\beta d_1)$$

$$100 z_0 + 100 z_0 \tan^2(\beta d) = z_0^2 + 10^4 \tan^2(\beta d)$$

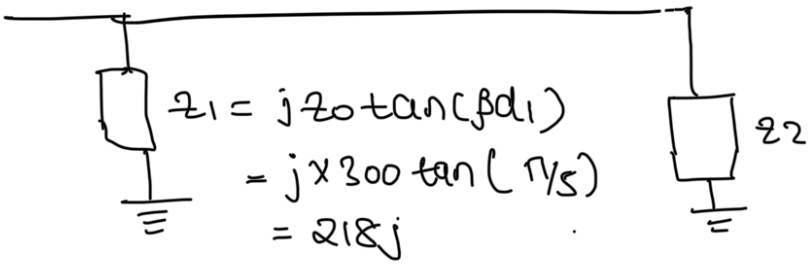
$$100 z_0 - z_0^2 = (10^4 - 100 z_0)\tan^2 \beta d$$

$$\frac{z_0}{100} = \tan^2 \beta d \Rightarrow \tan \beta d = \sqrt{2}$$

$\beta = 2\pi \Rightarrow d = 15.2\text{ cm}$

$d_1 = 9.8\text{ cm}$

(Q15) $\lambda = 100\text{ cm} \Rightarrow \beta = 2\pi \qquad z_0 = 300\ \Omega$

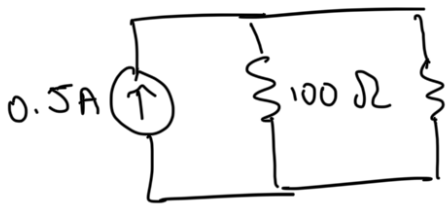


$$300 = \frac{z_1 z_2}{z_1 + z_2} \Rightarrow \frac{300 z_1}{z_1 - 300} = z_2 = 103.6 - 142.7j$$

$$z_2 = 300\left[\frac{z_L + j \cdot z_0 \tan(\pi/2)}{z_0 + j z_L \tan(\pi/2)}\right] = \frac{z_0^2}{z_L}$$

$$\Rightarrow z_L = \frac{9 \times 10^4}{103.6 - 142.7j} = 300 + 413j\ \Omega$$

Q18



$$Z = 50\left[\frac{25+50\tan(5.2\pi)j}{50+25\tan(5.2\pi)j}\right]$$

$$= 33.74 + 24.07j\ \Omega$$

$$100x(0.5-x) = (33.74+24.09j)x$$

$$\Rightarrow x = 0.36 - 0.07j$$

$100\ \Omega \rightarrow 0.14 + 0.07j \Rightarrow P = \frac{1}{2}i^2R = 1.225W$

$$V_S = 14 + 7j$$

$$P(Z) = Re\left(\frac{1}{2}I^2Z\right) = 2.269\ W$$

Q19 $Z_L = 100 + 50j \quad Z_0 = 50$

$$\Gamma_L = \frac{1+j}{1+3j} \Rightarrow |\Gamma|^2 = 20\% \text{ (reflected)}$$

$$\text{avg reflected power} = |\Gamma|^2\frac{V_0^2}{2Z_0} = \frac{1}{5\times100} = 2mW$$

Q20 $Z_L = 120\ \Omega \quad f = 6\times10^9$

$Z_g = 50\Omega \quad Z_0^2 = Z_g Z_L \Rightarrow Z_0 = 77.45\Omega$

$$\lambda = v/f = \frac{2.5\times10^8}{6\times10^9} = \frac{1}{24}m \Rightarrow l = \frac{\lambda}{4} = \frac{1}{96}m$$